

MARK SCHEME for the October/November 2008 question paper

9709 MATHEMATICS

9709/04

Paper 4, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2008 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



Page 2	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2008	9709	04

Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

Page 3	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2008	9709	04

The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only - often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR -1	A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR -2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA -1	This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

Page 4	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2008	9709	04

1	(i) (a) $10 - 8\cos\theta$	B1	
	(b) $8\sin\theta$	B1	[2]
	(ii)	M1	For using $X^2 + Y^2 = R^2$ or for using the cosine rule in the relevant triangle
	$(10 - 8\cos\theta)^2 + (8\sin\theta)^2 = 8^2$ or $10^2 + 8^2 - 2 \times 10 \times 8 \cos\theta = 8^2$	A1ft	
	$\cos\theta = 5/8$	A1	[3] AG
	First alternative for (ii)		
	$[\cos\phi = (10 - 8\cos\theta)/8 \text{ and } \sin\phi = 8\sin\theta/8]$	M1	For using $\cos\phi = X/R$ and $\sin\phi = Y/R$
	$8\cos\phi = (10 - 8\cos\theta)$ and $\phi = \theta$	A1ft	
	$\cos\theta = 5/8$	A1	AG
	Second alternative for (ii)		
	$[5, \sqrt{39}, 64]$	M1	For assuming $\cos\theta = 5/8$ and hence finding exact values of $\sin\theta$, X, Y and $X^2 + Y^2$
	$R = 8$	A1	
	→ assumption correct	A1	
	SR for (ii) (max 2/3)		
		M1	For assuming $\cos\theta = 5/8$ and hence finding $\theta = 51.3^\circ$ and the values of X, Y and $X^2 + Y^2$
	$R = 8 \text{ or } 8.0 \text{ or } 8.00 \text{ or } 7.997..$		
	→ assumption correct	A1	
2		M1	For resolving forces parallel to the plane (either case)
	$[R = 197, F = 63.0]$	M1	For using $F = 0.32R$ and $R = 20g\cos 10^\circ$ (or $20g\sin 10^\circ$ if this is part of a consistent sin/cos interchange)
	(i) $P = F + 20g\sin 10^\circ$	A1	
	Least magnitude is 97.8N	A1	

Page 5	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2008	9709	04

(ii) $P = F - 20g \sin 10^\circ$	A1ft	ft with $P \cos 10^\circ$ instead of P or sign error or cos instead of sin in component of weight	
Least magnitude is 28.3N	A1	[6]	
SR (for candidates who omit g) (max 3/6)			
For $P = F + 20 \sin 10^\circ$ in (i) and			
$P = F - 20 \sin 10^\circ$ in (ii)	B1		
	M1		For using $F = 0.32R$ and $R = 20 \cos 10^\circ$
Least magnitude is 9.78N in (i) and 2.83 in (ii)	A1		
3 (i)	M1		For applying Newton's second law (3 terms)
$F - 900 = 1200a$	A1		
$[18000/25 - 900 = 1200a]$	M1		For using $F = P/v$
Deceleration is 0.15ms^{-2}	A1	[4]	Accept $a = -0.15$
(ii) $18000/v - 900 = 0$	B1		
Least speed is 20ms^{-1}	B1	[2]	AG
4 $[s = (0 + 0.5)/2 \times 7]$	M1		For using $(u + v)/2 = s/t$
$s = 1.75 \text{m}$	A1		May be implied
PE gain = $160g \times 1.75$	B1ft		
KE gain = $\frac{1}{2} 160 \times 0.5^2$	B1		
$[WD = 2800 + 20]$	M1		For using $WD = \text{PE gain} + \text{KE gain}$
Work done is 2820J	A1	[6]	
SR (max 4/6) for candidates who use a non-energy method			
$[s = (0 + 0.5)/2 \times 7]$	M1		For using $(u + v)/2 = s/t$
$s = 1.75 \text{m}$	A1		
$[a = 1/14, T = 160g + 160/14, WD = 1611.4... \times 1.75]$	M1		For finding the acceleration and using Newton's second law (3 terms) to find the tension in the rope, then multiplying by the distance
Work done is 2820J	A1		

Page 6	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2008	9709	04

5 (i) (a)	$[5 = 0 + 2a]$	M1	For using $v = u + at$ with $u = 0$
	Acceleration is 2.5ms^{-2}	A1 [2]	
	(b)	M1	For applying Newton's second law to A (3 terms): (can be scored in (ii) by applying Newton's second law to B instead)
	$0.5g - T = 0.5 \times 2.5$	A1ft	
	Tension is 3.75N	A1 [3]	
(ii)	$T - mg = 2.5m$	B1ft	ft from $T - 0.5g = 0.5 \times 2.5$ in (i) (b) to allow $mg - T = 2.5m$ or $mg - 0.5g = 0.5 \times 2.5 + 2.5m$
	or $0.5g - mg = 0.5 \times 2.5 + 2.5m$		
	$[(10 + 2.5)m = 3.75]$	M1	For solving 3 term equation for m
	$m = 0.3$	A1 [3]	

6 (i)	- v is single valued, continuous and positive for $0 < t < 1000$. 1st segment has +ve slope	M1	For sketching a graph consisting of 3 straight line segments, for which (see left):
	And two or more of $v(0) = 0$ $v(1000) = 0$ 2nd segment has zero slope 3rd segment has -ve slope		
	Correct sketch	A1 [2]	
(ii)		M1	For using 'area under graph' represents distance of 20000m
	$\frac{1}{2} (600 + 1000)V = 20\,000$	A1	
	$V = 25$	A1 [3]	
SR for candidates who assume 1 st and 3 rd time intervals are each 200 s (max 2/3)			
	$\frac{1}{2} V \times 200 + V \times 600 + \frac{1}{2} V \times 200 = 20000$	B1	
	$V = 25$	B1	

Page 7	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2008	9709	04

(iii) $[V/t_1 = 0.15 \rightarrow t_1 = 166.6\dots]$	M1	For using the gradient property for acceleration (or $v = 0 + at$) to find t_1 .	
$t_3 = 233.3\dots$	A1ft	ft $400 - V/0.15$	
$[s_3 = \frac{1}{2} 233.3\dots \times 25]$	DM1	For using the area property for distance or $(u + v)/2 = s/t$. Depends on previous M1	
Distance is 2920m	A1	[4]	
		Alternatively	
		For using $V^2 = 2 \times 0.15 s_1$	M1
		$[\rightarrow s_1 = 2083.3\dots]$	
		$s_2 = 15000$	B1ft
		(ft 600V)	
		For $s_3 = 20000 - s_1 - s_2$	DM1
		Distance is 2920m	A1

7 (i) $[v^2 = 2 \times 10 \times 1.25 \text{ or } \frac{1}{2} mv^2 = mg(1.25),$	M1	For using $s = 1.25$ and $a = 10$ to	
$1.25 = \frac{1}{2} 10t^2]$		find either v or t	
Speed of P is 5ms^{-1}	A1		
Time taken is 0.5s	A1	[3]	
(ii)	M1	For using $v = \int a(t)dt$	
$v = 10t - 0.15t^2 \quad (+C)$	A1		
$v = 10t - 0.15t^2 + 5$	A1ft	ft wrong answer in (i)	
	M1	For using $x = \int v(t)dt$	
$x = 5t^2 - 0.05t^3 + 5t$	A1ft		
$[x = 5 \times 2.5^2 - 0.05 \times 2.5^3 + 5 \times 2.5 \quad (= 42.97)]$	DM1	For substituting $t = 3 - t_1$ in $x(t)$. Depends on both previous M1s.	
Distance OP is 44.2m	A1	[7]	